

Pooja Kiran

Security Engineer | Agent Security | Application Security | Cloud IAM
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EXPERIENCE

Independent AI Security Researcher & Engineer

Aug. 2024 - Present

Self-Directed Research

Tempe, AZ

- **2024:** Built the OS Resource Management Simulator, a Flask/Docker IFT 510 lab, then evolved the design into a PostgreSQL lease service with atomic acquisition, expiring leases, idempotent retries, role-based API access, audit history, metrics, and concurrency tests.
- **2025:** Applied AWS IAM roles and policy evaluation, least-privilege design, CloudTrail logging, and compliance controls in graduate security work, establishing the cloud identity foundation for later security tooling.
- **2026:** Built and evaluated security tooling across agent runtime enforcement, AWS IAM analysis, model supply-chain scanning, and training-data poisoning detection; standardized evidence with CI, SARIF/Code Scanning, Elastic Security, Prometheus, Docker, and reproducible tests.

Business & Compliance Lead, AEROSSEC

Aug. 2025 - Dec. 2025

Honeywell Aerospace Technologies × Arizona State University

Tempe, AZ

- Led business/compliance analysis for a proposed airline PSS protection layer and built a **\$120K first-year commercialization scenario** with a five-year financial model presented to ASU and Honeywell stakeholders.

Graduate Teaching Assistant, IT Grader

Jan. 2025 - Oct. 2025

Ira A. Fulton Schools of Engineering, Arizona State University

Mesa, AZ

- Evaluated approximately **85 undergraduate web-development submissions per semester** and **52 graduate cybersecurity assignments**, giving evidence-based feedback on security controls, policy, compliance, risk analysis, and implementation quality.

PROJECTS

MCP Agent Security Gateway | *Python, FastAPI, MCP/JSON-RPC 2.0, Elastic Security*

Jul. 2026 - Present

- Engineered an inline policy gateway with default-deny capability checks, **55 prompt-injection patterns**, PII/exfiltration and anti-SSRF controls, rate limiting, fail-closed enforcement, and tamper-evident audit logs; verified **718 tests at 82.46% coverage**, **9 Elastic rules**, and **21 SIEM tests**.

AWS Agent Identity Guard | *Python, AWS IAM, SARIF 2.1.0*

Aug. 2026 - Sep. 2026

- Built a static IAM analyzer with **25 deterministic rules** for wildcard grants, `iam:PassRole`, `sts:AssumeRole`, privilege escalation, trust-policy weaknesses, audit tampering, and permission boundaries; verified **235 passing tests** plus SARIF 2.1.0 and GitHub Code Scanning output.

HF Model Provenance Scanner | *Python, SafeTensors, GGUF, ONNX, Keras*

Jul. 2026 - Sep. 2026

- Developed a non-executing model supply-chain scanner using pickle-opcode, AST/taint, dependency/provenance, obfuscation, and binary-format checks; verified **241 tests at 75.81% coverage** and detected **33/33 committed adversarial fixtures**.

Dataset Poisoning Detector | *Python, FastAPI, Kafka, Redis, Prometheus*

Jul. 2026 - Present

- Built statistical and spectral training-data screening with Isolation Forest, label-aware analysis, drift detection, deduplication, streaming, and quarantine; verified **199 tests at 90.88% coverage** and **0.55 / 0.60 / 0.70 F1** at 5% / 10% / 20% poisoning.

EDUCATION

Arizona State University

Tempe, AZ

M.S., Information Technology (Security), GPA: 3.87/4.00

Aug. 2024 - May 2026

M. S. Ramaiah University of Applied Sciences

Bengaluru, India

B.Tech., Computer Science & Engineering, CGPA: 8.44/10

Aug. 2019 - Aug. 2023

TECHNICAL SKILLS

Agent & AI Security: MCP/JSON-RPC 2.0, Tool/Function-Call Security, Prompt Injection, LLM Red Teaming, Model Supply-Chain Security, Training-Data Integrity, Adversarial ML

Cloud & Application Security: AWS IAM, Least Privilege, Trust Policies, Permission Boundaries, Threat Modeling, API Security, Capability-Based Authorization, Data Exfiltration Detection

Engineering & Detection: Python, Rust, C++, FastAPI, pytest, Hypothesis, GitHub Actions, SARIF 2.1.0, CodeQL, Docker, Kubernetes, Elastic Security, Prometheus

CERTIFICATIONS

AWS Academy Graduate - Cloud Architecting (Apr. 2025) | AWS Academy Graduate - Cloud Security Foundations (Nov. 2025)